

Industrial engineering solutions enhance operational efficiency in human and veterinary hospitals: a scoping review

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Abstract

Human and veterinary healthcare systems face many parallel challenges, yet opportunities for cross-sectoral learning remain underexplored. This scoping review aimed to compare healthcare delivery in both domains, highlighting operational inefficiencies, shared challenges, and innovative strategies that may enhance outcomes in both sectors. A search and review of 43,987 articles was conducted across multiple databases. After applying inclusion and exclusion criteria and conducting quality assessments, 100 studies were selected for detailed analysis. The review focused on operational elements such as patient scheduling, staffing, electronic health records, patient flow, satisfaction, wait times, length of stay, facility layout, and resource allocation. The review identified recurring inefficiencies in both sectors, notably in scheduling and resource planning. Common operational challenges in healthcare include poor electronic health records integration, suboptimal facility layout, and understaffing. Innovative solutions like improved protocols, Lean Six Sigma, smart scheduling, artificial intelligence, and ergonomics can improve efficiency and patient outcomes. The review findings underscored the potential for shared solutions between human and veterinary healthcare systems. Recognizing and applying successful strategies across both fields could improve operational efficiency, patient satisfaction, and overall quality of care. Veterinary healthcare lags in adopting industrial engineering solutions to solve operational challenges. This review provided invaluable information from the literature and could serve as a blueprint in multiple areas to expedite improvements in veterinary care. Human healthcare systems are much more standardized than veterinary systems and provide multiple potential solutions to improve veterinary care delivery.

Keywords: industrial engineering, human hospitals, veterinary hospitals, healthcare systems, operational inefficiencies

The US healthcare system is a complex, multilayered operational realities. Like human healthcare, veterinary network comprising dynamic components and sub-nary healthcare is supported by a network of private components. It begins with patients requiring healthcare services provided by healthcare workers. These services are offered in various settings, including hospitals and clinics, funded by federal and state programs, private investors, and healthcare insurance providers. All these elements are interconnected through technology that keeps the system operational.^{1,2}

Similarly, veterinary healthcare in the US plays a crucial role in maintaining both animal and public health. Veterinary services are essential for ensuring the health and well-being of pets, livestock, and wildlife. This sector includes veterinary clinics, animal hospitals, and research institutions, each with distinct

investors, government programs, and insurance providers and relies heavily on advanced technologies for accurate record keeping, diagnosis of disease, and the treatment of complicated medical conditions.^{2,3}

Pets, considered family members in many households, receive preventative, emergency, routine, and specialty care from veterinary professionals, enhancing their quality of life and longevity and improving the emotional well-being of their owners.²⁻⁴ Comparative studies between human and veterinary healthcare systems can identify common challenges and innovative solutions that benefit both fields, as represented in **Table 1**.

By examining operational aspects such as scheduling, staffing, and the use of electronic health records (EHRs), we can develop strategies to improve efficiency and outcomes across both sectors. Understanding the parallels and differences in

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Table 1—Comparison of human and veterinary healthcare systems.

Aspect	Similarities	Differences
Funding sources ⁵⁻⁷	A mix of private and public sources	Fewer government funding sources for veterinary
Care providers ^{3,8,9}	Highly trained professionals	A broader range of species-specific expertise in veterinary
Facilities ^{2,3}	Dedicated care facilities	More common mobile services in veterinary
Operational challenges ^{8,10}	Similar operational challenges	More resource constraints and smaller team sizes in veterinary
Patient flow and scheduling ^{8,9}	Need efficient flow and scheduling	More complex patient flows in human healthcare
Record keeping ^{11,12}	Digital record-keeping	Stricter EHR regulations in human healthcare
Ergonomics ^{8,13}	Aim to reduce strain, improve safety	Focus on species-specific ergonomics in veterinary
Impact of facility layout ^{14,15}	Importance of layout for efficiency/safety	More focus on infection control in human healthcare

Comparison of human and veterinary healthcare systems across funding, care providers, facilities, operational challenges, patient flow, record keeping, ergonomics, and facility layout is summarized with similarities and differences.

EHR = Electronic health record.

facility layout, ergonomics, resource allocation, and capacity planning can lead to cross-sectoral learning and advancements in healthcare delivery.

The objective of this review was to address the operational challenges faced by human and veterinary hospitals by bridging the gap between the 2 healthcare systems. Veterinary healthcare would benefit immensely from adapting human healthcare's institutionalized best practices that have been tested and validated.

This comprehensive review aimed to ensure that both human and veterinary healthcare systems are equipped to meet evolving demands and continue providing high-quality, inclusive care, contributing to the broader One Health paradigm and advancing family equity in care systems.^{3,16,17}

Methods

In our research, we employed a comprehensive literature search strategy, represented in **Figure 1**, by leveraging the following databases: Google Scholar, CAB Abstract, PubMed, IEEE Xplore, and

ScienceDirect. The most recent search of all databases was conducted in April 2025. These platforms

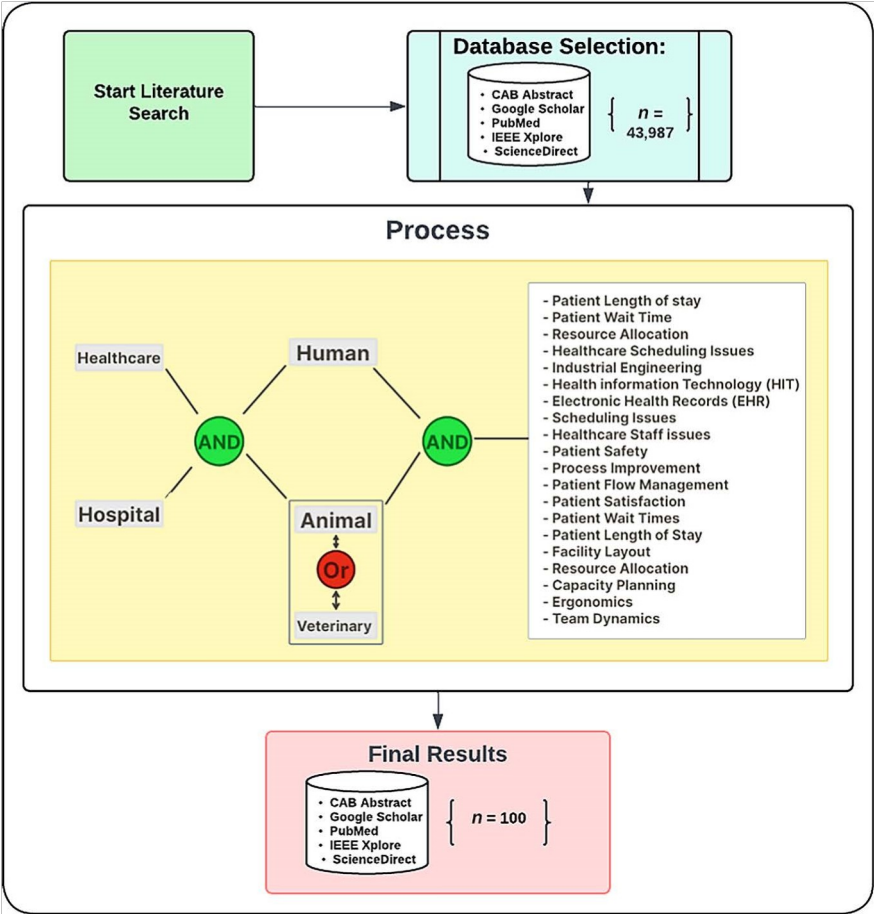


Figure 1—Literature search strategy and selection process. This diagram illustrates the literature search conducted across 5 databases (CAB Abstract, Google Scholar, PubMed, IEEE Xplore, and ScienceDirect), yielding 43,987 initial articles. Search terms combined healthcare, hospitals, and human medicine with veterinary/animal medicine using Boolean operators (AND/OR), as shown in the flow diagram. Key operational topics included scheduling, resource allocation, patient satisfaction, patient safety, health information technology, ergonomics, and team dynamics. After applying inclusion and exclusion criteria, 100 peer-reviewed studies were retained for detailed analysis.

were chosen because they encompass a broad spectrum of medical and engineering literature, which is vital for our interdisciplinary study. Notably, the CAB Abstract database housed the majority of papers relevant to the veterinary aspect of our research. This database proved to be a rich source of information, contributing significantly to our study. The search used a set of specifically tailored keywords to capture the full range of topics, as indicated.

Our inclusion criteria encompassed peer-reviewed articles published from 2015 onward, focusing on both human and veterinary healthcare systems and explicitly addressing the application of industrial engineering methods in healthcare. Foundational papers published before 2015 were included only if (1) no recent publications addressed the same concept, or (2) the earlier work remained highly cited and central to the field. Exclusion criteria included (1) non-peer-reviewed articles, (2) papers published before 2015 that did not meet the above inclusion triggers, (3) articles not available in full text, (4) studies not written in English, (5) studies outside of healthcare (eg, unrelated industries), and (6) papers lacking sufficient methodological detail to evaluate relevance.

The process of selecting and evaluating relevant studies was rigorous; it began with screening of titles and abstracts to determine their relevance to our research questions. Articles that passed this initial phase were then subjected to a full-text review, where their suitability was assessed based on more detailed criteria such as study design, relevance of industrial engineering applications, and focus on healthcare improvements. The final selection of studies underwent a quality assessment using standardized tools appropriate for each study type.

In synthesizing and analyzing the findings, we extracted vital information from each study, including study aims, methodologies, outcomes, and critical findings relevant to process improvements in healthcare. A comparative analysis was then conducted to identify common themes, differences, and outcomes between human and veterinary healthcare systems. The industrial engineering perspective focused on analyzing how various methodologies have facilitated process improvements and quality enhancements in healthcare settings.

Results

Our search and selection process identified 43,987 articles through the database search. After applying the inclusion and exclusion criteria, 100 studies were retained for comparative analysis. Studies were excluded during screening for several reasons, including failure to meet inclusion criteria, duplication, or inadequate quality. These steps ensured that only studies meeting the required standards were included in the final analysis.

Scheduling issues

Scheduling remains a critical yet complex component of healthcare management, directly influencing efficiency, resource utilization, and patient satisfaction. Hospitals must balance elective and emergency procedures, optimize operating room use, and manage staff workloads amid unpredictable conditions. Industrial engineering provides powerful tools such as mixed-integer linear programming, simulation, and heuristic methods to address these

challenges as presented.¹⁸ Case studies^{19,20} illustrate the value of these approaches: decision support systems and scheduling algorithms have reduced wait times, improved operating room utilization, and minimized operational costs.

For instance, applied generalized disjunctive programming was used to boost surgical throughput and optimize appointment scheduling, while mixed-integer linear programming and heuristics were applied to reduce surgery days and improve operating room allocation.²¹

Veterinary healthcare faces similar constraints, particularly due to the limited translatable research, and introduces distinct challenges. For example, veterinary teams often manage multiple patients simultaneously, with fewer specialized staff and limited equipment resources compared to human healthcare. These factors demand real-time flexibility, as inefficiencies in scheduling can create cascading delays and increase costs, underscoring the need for tailored systems that reflect veterinary-specific operational realities. These insights emphasize that both human and veterinary sectors benefit from dynamic, data-driven scheduling approaches customized to their unique demands.²²

Healthcare staff issues and patient safety

Staff well-being and patient safety are closely connected in both human and veterinary healthcare. Burnout, fatigue, and operational stress negatively impact care quality, making comprehensive strategies essential. Incorporating well-being metrics into performance dashboards enables real-time monitoring and proactive intervention.⁸ Predictive analytics can identify burnout trends early, allowing for preventative measures before patient safety is compromised.¹ Additionally, optimizing workflows, offering resilience training, and adjusting staffing models to balance workloads help reduce stress and enhance care quality.⁸

Veterinary healthcare faces additional challenges due to smaller team sizes and limited resources. Establishing community support systems and mental health partnerships is critical to supporting staff.^{10,23} Technologies like practice information management systems (PIMS) and telemedicine streamline tasks and reduce administrative burdens, improving both staff well-being and patient safety.^{23,24}

Fatigue from long hours is a major risk factor for burnout and can potentially compromise patient safety, especially in veterinary care, where insufficient rest leads to performance declines and increased errors.²⁵ Addressing this through flexible schedules, better shift planning, and peer support can greatly improve outcomes.^{10,23} Promoting a “culture of safety” with open communication, incident reporting, and nonpunitive feedback further supports staff and elevates overall care quality.^{23,26}

While scheduling inefficiencies and workload distributions are important contributors to staff fatigue, it is equally critical to acknowledge the psychological stressors unique to veterinary medicine. Factors such as frequent exposure to euthanasia, emotionally charged client interactions, and compassion fatigue also play a major role in burnout. Addressing these requires not only operational improvements but also access to mental health

resources, peersupport initiatives, and resilience-building programs tailored to the realities of veterinary practice, including financial pressures such as the cost of care and owner funding constraints.

Health information technology and EHRs

Health information technology (HIT) plays a vital role in addressing inefficiencies in both human and veterinary healthcare. In 2023, US healthcare expenditures totaled \$4.9 trillion, projected to reach 19.7% of gross domestic product by 2032.^{27,28} In comparison, veterinary healthcare

In human healthcare, Lean principles have successfully improved patient flow, reduced wait times, and increased efficiency across various settings.³⁸ Applications in inpatient wards have led to better resource utilization, while combining Lean with DES in outpatient care improves scheduling and resource allocation.^{39,40} These integrated approaches significantly enhance operational performance and patient satisfaction. Adapting these strategies to the unique characteristics of healthcare settings is key to maximizing their impact.⁹

Veterinary healthcare has also benefited from process improvement strategies. Surgical checklists improve

Table 2—Comparison of human and veterinary electronic medical systems.

Feature	Human EMRs	Veterinary PIMS
Billing and finance ^{33–36}	Native capability	Native capability
Reminders and tracking procedures ^{33–36}	Robust system for tracking patient care	Less sophisticated system for tracking patient care
Medical notes ^{33–36}	Supports both free text and structured medical notes	Supports free text medical notes and limited support for structured notes
Decision support systems ^{33–36}	Native capability	Not available
Laboratory and imaging integration ^{33–36}	Integrates with LIMS, PACS, and RIS	Generally, not available
Order entry and adverse event detection ^{33–36}	Facilitates computer physician order entry and detects adverse events	Not available

Comparison of human and veterinary electronic medical systems and key features including billing, reminders, notes, decision support, laboratory integration, and order entry are presented with similarities and differences.

EMRs = Electronic medical records. LIMS = Laboratory information management systems. PACS = Picture archiving and communication systems. PIMS = Practice information management systems. RIS = Radiology information systems.

expenditures totaled \$151.9 billion in 2024 with a projected \$157 billion in 2025.^{29,30} These significant investments underscore the importance of technology in improving care coordination, delivery, and administration.^{3,8} Adopting EHRs in human healthcare involves high initial costs and extensive training but leads to improved care quality and reduced errors, supported by initiatives like the Health Information Technology for Economic and Clinical Health Act.^{31,32} Administrative data systems require limited interoperability, whereas clinical data demand high interoperability and strict adherence to Health Insurance Portability and Accountability Act standards.^{8,31} Veterinary healthcare faces unique challenges in HIT adoption. High costs, lack of standardization, and the need for significant workflow adaptation especially for smaller practices limit progress.^{3,4,8,11,16} Veterinary PIMS often focus on billing and recordkeeping rather than full interoperability, hindering advanced analytics and patient care.^{3,11} With an estimated 495 million clinical records generated annually, scalable and flexible systems are critical. As outlined in **Table 2**, human EMRs are more robust due to regulatory demands, while veterinary PIMS remain operationally focused.³⁷ Bridging this gap through cross-sectoral learning could enhance veterinary care and practice management.

Process improvement and patient flow management

Process improvement and efficient patient flow are essential for delivering high-quality, timely care in both human and veterinary healthcare. Methodologies such as Lean, Six Sigma, and discreteevent simulation (DES) have been widely adopted to optimize operations and enhance patient outcomes.

workflow, reduce errors, and enhance postoperative care, minimizing the need for follow-up visits.⁴¹ The broader use of safety checklists aligns with human healthcare practices, optimizing resource use and reducing treatment delays.⁴²

The synergy of Lean and DES is particularly effective in addressing operational challenges.⁴⁰ While most research emphasizes human healthcare, these methodologies translate well to veterinary settings, offering improvements in flow management, cost reduction, and patient and client satisfaction.^{41,42} Cross-sectoral learning supports the adoption of proven strategies to elevate efficiency and care quality across both fields.

Patient satisfaction

Achieving high patient satisfaction is essential for evaluating care quality and operational effectiveness in healthcare. Numerous studies point to inefficiencies such as long wait times and redundant procedures as major contributors to poor patient experiences. Sedlack et al⁴³ found that prolonged processing times in the Emergency Department’s Pink Pod at Advent Health stemmed from systemic bottlenecks. Similarly, Nino et al⁴⁴ identified excessive delays and duplications in the central registration process at a Montana healthcare facility. Bordoloi⁴⁵ documented inefficiencies in phone service operations, where patients experienced long call wait times, unreturned messages, and frequent transfers.

Industrial engineering tools have been effectively applied to resolve these issues. Lean Six Sigma methodologies tools like Value Stream Mapping, Fishbone diagrams, and Pareto charts identify and eliminate waste, streamline processes, and reduce patient wait times.^{43,44}

Queuing models were used to optimize call center staffing, enhancing response times and overall service delivery.⁴⁵

Understanding patient expectations also plays a critical role in satisfaction. Game theory-based models were utilized to predict patient choices in China's hierarchical healthcare system, offering insights into patient behavior and improving service alignment.⁴⁶ The Kano model helped to classify hospital service attributes, enabling targeted improvements in areas most valued by patients.⁴⁷

Collectively, these studies demonstrate that applying industrial engineering and behavioral analysis tools leads to measurable improvements in patient satisfaction. Reductions in complaints, increased operational efficiency, and better alignment with patient expectations all underscore the value of these methodologies in modern healthcare management.

Staffing levels are a critical factor in patient satisfaction across both human and veterinary healthcare. Adequate staffing reduces wait times, improves care quality, and prevents burnout, which is closely linked to decreased patient safety and satisfaction.^{10,24,48} Similarly, burnout among veterinarians and technicians compromises care quality and increases error rates.⁴¹

The EHRs present a mixed impact on satisfaction. While they enhance data accuracy and care coordination, their administrative demands can lead to staff burnout and reduced face-to-face patient interaction.^{41,49} This is true in both sectors. However, patient-generated health data can improve engagement when integrated thoughtfully.^{12,50}

In human healthcare, satisfaction is typically measured from the patient's perspective, whereas in veterinary healthcare, it is assessed through client satisfaction, reflecting the perceptions of pet or animal owners. Although many of the same operational factors such as wait times, facility design, and staff availability affect all healthcare, veterinary client satisfaction is also strongly influenced by communication quality, cost transparency, and the emotional nature of care decisions, including euthanasia.

Dissatisfaction at the client level can directly affect staff well-being, as negative client interactions have been identified as a significant driver of stress and burnout in veterinary healthcare teams. Scheduling inefficiencies may contribute to frustration, but broader psychosocial factors such as compassion fatigue, client expectations, and emotionally demanding encounters play an equally important role in shaping satisfaction and staff resilience.

Efficient patient process flow is equally important. Poor workflow contributes to longer wait times and reduced provider interaction, negatively impacting satisfaction.⁵¹

Optimizing patient flow in both human and veterinary healthcare improves client experiences and reduces stress for patients and animals alike.^{41,42} Staff involvement in workflow redesign leads to more effective patient management and satisfaction.⁵²

Facility layout and ergonomics also play a significant role. Human healthcare benefits from well-designed spaces that reduce staff strain and enhance patient experiences.²⁵ Veterinary facilities with ergonomic features like noise control and adjustable tables improve staff performance and animal welfare, enhancing client satisfaction.⁵³

In both sectors, each topic can significantly boost satisfaction leading to a better overall care quality.

Patient weight times

Patient wait times are a critical indicator of healthcare service quality and operational efficiency. Prolonged wait times often reflect deeper issues such as scheduling mismanagement, operational bottlenecks, and resource constraints.⁵⁴⁻⁵⁶

These inefficiencies, especially in specialized units like breast imaging and head and neck cancer clinics, contribute significantly to reduced patient satisfaction.^{57,58} Scheduling variability and poor prioritization further exacerbate these challenges.^{59,60}

Industrial engineering techniques such as Lean methodologies, DES, and advanced scheduling offer proven solutions to streamline processes and enhance patient flow.^{61,62} Hospitals have achieved over a 30% reduction in wait times through optimized resource use and demonstrated sustained improvements in satisfaction scores through continuous process enhancements.^{58,63}

Innovative interventions across various departments further validate these strategies, help improve patient and staff satisfaction, and reduce wait times through refined scheduling. Huang⁶⁴ introduced flexible appointment slots to significantly cut delays.

Veterinary healthcare faces parallel issues. A study⁶⁵ in southwest Nigeria highlighted how equipment shortages and understaffing in veterinary laboratories lead to significant delays. Veterinary care involves additional diagnostic and handling complexities, which amplify scheduling challenges in under-resourced settings.

In veterinary settings, however, client satisfaction is often most directly influenced by the time it takes to see a veterinarian rather than delays in diagnostic testing. Extended waiting to consult with the doctor can heighten client frustration, particularly in cases involving distressed or anxious animals. While diagnostic delays play a role, timely access to the veterinarian remains the most critical factor shaping client perceptions of wait times and overall service quality.

Collectively, these studies illustrate that applying industrial engineering methods can substantially reduce impatient wait times in both human and veterinary healthcare. Streamlined operations, strategic scheduling, and efficient resource allocation are vital to improving care delivery and patient outcomes.

Patient length of stay

Extended length of stay (LOS) in hospitals, particularly in emergency departments, is a critical challenge that impacts patient satisfaction, clinical outcomes, and operational efficiency.⁶⁶ Causes often include patient flow inefficiencies, resource mismanagement, and procedural bottlenecks. Addressing these issues requires innovative approaches grounded in industrial engineering and operations research, including simulation, predictive modeling, and process improvement techniques.⁶⁷

Numerous studies demonstrate the effectiveness of such methodologies, which are shown in **Table 3**.⁶⁸ Lean Six Sigma (define, measure, analyze, improve, control) was used to reduce LOS and improve patient satisfaction, while

Fuzzy Logic was applied to optimize patient flow.^{69,70} The extended Deming Cycle was utilized to streamline admission/discharge processes, and integrated Lean

systematic layout planning with human reliability analysis (human error/ risk assessment) and fuzzy analytic hierarchy process (decision-making framework) has been shown to

Table 3—Overview of industrial engineering methods to optimize patient length of stay in healthcare settings.

Problem focus	Methodology used	Key results
Inefficiencies in patient flow and process bottlenecks ⁶⁸	Lean Six Sigma DMAIC methodology	30% reduction in LOS, a significant increase in patient satisfaction
High variability in patient arrivals and complex ED ⁶⁹	Fuzzy logic and CPN processes	Improved patient processing rates and reduced LOS
Inefficiencies in the admission-discharge process ⁷⁰	Extending the Deming Cycle method	Reduced emergency department boarding times and cost savings
Variability in patient arrival and ED operations ⁶⁷	DES	Optimized staffing levels and reduced patient LOS
Inadequate resource allocation and inefficiencies in patient processing ⁷¹	Lean principles with DES	Enhanced patient flow and improved operational efficiency
Difficulties in predicting patient throughput times ⁷²	ANNs	Enhanced capability to predict LOS with greater accuracy
Predicting LOS for Dengue patients during epidemics ⁷³	Data mining with the C4.5 algorithm	Accurate resource planning during peak demand improved LOS prediction
Variability of patient inflow and dynamic conditions in ED ⁷⁴	Observational study with Ishikawa causal diagram	Identified key organizational factors affecting LOS and suggested improvements in scheduling and communication
Management of patient flow amid varying arrival patterns ⁷⁵	Retrospective study with statistical analyses	Identified operational interventions to enhance emergency care timeliness
Impact of patient volume and complexity of cases on LOS ⁷⁶	Retrospective observational study with multiregression models	Found significant factors affecting LOS, offered insights for better patient flow and resource management

Industrial engineering methods used to optimize patient length of stay (LOS) in healthcare settings are shown. Methods include Lean Six Sigma, fuzzy logic, discrete-event simulation, neural networks, and regression models. Key results are summarized.

ANNs = Artificial neural networks. CPN = Colored petri net. DES = Discrete event simulation. DMAIC = Define, measure, analyze, improve, and control. ED = Emergency department.

principles with DES aided in improving resource allocation and reducing wait times. Artificial neural networks were pivotal for LOS prediction, while data mining with the C4.5 algorithm has proven effective for better resource planning.^{67,71,73}

Further studies examined organizational factors using observational and statistical methods,⁷⁶ applied multiregression analysis to identify influences on LOS, and advised interventions to improve emergency care timeliness and long-term LOS outcomes.^{63,74}

In veterinary medicine, LOS is not usually a central performance metric outside of intensive care or hospitalization. Most veterinary patients are discharged on the same day of their visit or procedure. However, in cases involving hospitalization or complex surgeries, extended LOS may occur due to illness severity, complications, recovery monitoring, or limited staffing resources. While not directly equivalent to human hospital stays, these instances highlight how resource constraints and clinical complexity can prolong care in veterinary settings.⁷⁷

Facility layout

Facility layout plays a vital role in both human and veterinary healthcare, directly influencing operational efficiency, safety, and patient outcomes. In human healthcare, strategic layout planning addresses the complexity of hospital environments. Methods such as systematic layout planning (structured facility layout method), simulation-based optimization, and multicriteria decision-making are commonly used. Integrating

enhance logistics, space use, and safety in operating theaters.^{14,78}

Simulation techniques, including spatial network analysis and workflow modeling, quantify factors like visibility, walkability, and congestion to improve layouts. These approaches reduce stay_in_place distances, wait times, and crowding, improving staff workflow and patient satisfaction. Algorithms such as the Ant agent algorithm support adaptive layout design, enhancing resource utilization in outpatient clinics.^{79,80}

Veterinary healthcare shares similar goals, with a focus on animal welfare, logistics, and efficient care delivery. Simulation and optimization methods are employed to design layouts that balance operational constraints with species-specific care needs.⁸¹

Automated layout algorithms further enhance workflow efficiency and reduce operational costs while ensuring proper environmental conditions for animals.⁸²

Across both sectors, facility layout is key to optimizing care environments. Integrating advanced tools like virtual reality and promoting cross-sectoral learning can further improve layout designs and adaptability in evolving healthcare systems.

Resource allocation and capacity planning

Effective resource allocation and capacity planning are central to operational efficiency and high-quality care across healthcare systems. From emergency departments

to outpatient clinics and whole hospitals, healthcare providers must navigate unpredictable patient inflow and variable resource availability. Emergency departments commonly face overcrowding due to inconsistent patient volumes and acuity levels. Outpatient clinics and specialty services also struggle with irregular appointment demand and specialist availability, emphasizing the complexity of managing bed occupancy and interdepartmental coordination.^{83,84}

Researchers have applied advanced methodologies to address these challenges. Techniques include simulation optimization with metaheuristics, mixed integer programming with DES, and adaptive scheduling using real-time data.^{83,84} Hospital-wide studies incorporate forecasting and systemic optimization models to balance resources and reduce costs.

These interventions have demonstrated reductions in wait times, improved staff and bed utilization, and enhanced overall care delivery and have shown how such models perform effectively during crises, proving their scalability in emergencies.⁸⁵

In veterinary healthcare, limited resources such as equipment, space, and staffing magnify these challenges. Applying resource allocation theory and adopting human-style staffing strategies can improve capacity management. Collaborative models, successful during COVID-19 in human healthcare, could similarly strengthen veterinary hospitals during peak demand or emergencies.⁸⁶

Ergonomics in human and animal healthcare

Ergonomics is essential for enhancing safety, efficiency, and comfort in both human and animal healthcare settings. In human healthcare, ergonomic design addresses musculoskeletal disorders, commonly caused by long hours and physical strain from patient handling. Interventions such as adjustable workstations, adequate lighting, and user-friendly equipment reduce injuries and improve worker satisfaction and productivity.⁸⁷ The Systems Engineering Initiative for Patient Safety model links ergonomic principles to patient safety and care quality.²⁵ Ergonomic medical devices like height-adjustable beds and mobile carts further reduce physical strain and increase care efficiency.⁸⁷

In veterinary healthcare, ergonomics reduces physical strain on staff and enhances animal welfare. Tools like adjustable tables, ergonomic instruments, and strategically placed supplies minimize awkward movements and repetitive stress.⁸⁷ Environmental factors such as ventilation, lighting, and noise control are crucial for reducing animal stress.⁸⁸

Both sectors benefit from systematic ergonomic planning using simulation-based tools and decisionmaking frameworks. Technologies like virtual reality are applied to improve ergonomic designs.⁸⁹ Human healthcare emphasizes cognitive ergonomics and tech integration, while veterinary ergonomics focuses on physical handling and species-specific needs.⁸⁷ Cross-sector opportunities include adapting mechanical lifting devices and workstation ergonomics from human healthcare to veterinary settings.⁸⁷

Ultimately, ergonomic interventions improve safety, care quality, and operational efficiency in both sectors,

highlighting the value of cross-sectoral learning and collaboration.

Hospital team dynamics and their impact on healthcare outcomes

Healthcare teams operate in complex environments where effective collaboration is essential. Structured teams with clear roles, regular meetings, and mutual respect for diverse skills can reduce stress, enhance job satisfaction, and improve staff retention. These dynamics also lead to better patient care through improved communication and teamwork, all of which are reinforced by strong leadership.^{90,91} Research highlights that effective team functioning, supported by leadership, is strongly linked to improved patient outcomes.⁹⁰

In veterinary healthcare, the same principles apply. Defined team roles, routine meetings, and skill recognition contribute to improved care quality, fewer errors, and enhanced staff well-being.^{92,93} When veterinary teams are well-organized and supported by leadership, staff report lower stress and greater job satisfaction, which translates to higher performance and better patient care. Leadership is central in cultivating a cohesive team environment and enabling sound decision-making.⁹¹

Ultimately, both human and veterinary healthcare benefit from strategic team design and leadership and effective teamwork across both sectors consistently improves patient outcomes while also reducing workplace stress and increasing job satisfaction.

Discussion

This comparative study on operational challenges and solutions in human and veterinary hospitals has unveiled novel insights into optimizing healthcare systems through industrial engineering techniques. By delving into scheduling issues, patient satisfaction, patient wait times, patient LOS, resource allocation, and capacity planning, we have unearthed unique challenges and practical strategies that can be universally applied. Our analysis demonstrated that advanced scheduling systems reduce wait times and improve overall patient flow, directly enhancing patient satisfaction. Specifically, implementing mixed-integer linear programming and simulation models in scheduling significantly enhanced operational efficiency by optimizing resource allocation and reducing patient LOS. These results underline the critical role of precise and dynamic scheduling in managing healthcare facilities effectively.

Both human and veterinary healthcare systems face the perennial challenge of balancing resource constraints with the need for high-quality patient care. Our study highlighted how industrial engineering solutions can address these challenges, particularly in resource allocation and capacity planning. By employing discrete event simulation and capacity planning models, facilities can anticipate demand fluctuations and allocate resources more effectively, minimizing bottlenecks and enhancing service delivery. Feasibility, however, varies by practice size and resources, particularly in smaller veterinary clinics.

While our findings are promising, it is essential to acknowledge the limitations of our study. The variability in healthcare settings, such as differences in hospital sizes and available technologies, may impact the generalizability of

our results. This underscores the necessity for future research to focus on tailoring these industrial engineering techniques to diverse healthcare environments and exploring the integration of emerging technologies. Additionally, more longitudinal studies are required to assess the long-term impacts of these optimizations on patient outcomes and system sustainability, highlighting the ongoing relevance and potential for development in this field.

The implementation of HIT, process improvement strategies, ergonomic designs, and team dynamics is pivotal in enhancing healthcare outcomes, employee well-being, and operational efficiency across both human and veterinary settings. Effective HIT adoption enhances patient care quality, reduces errors, and streamlines operations.^{31,32} Lean and Six Sigma methodologies improve patient flow and optimize resource utilization, while ergonomic designs reduce physical strain and enhance safety for healthcare workers.^{38,94} Additionally, wellstructured team dynamics and strong leadership play crucial roles in improving job satisfaction, reducing stress, and boosting patient care.^{90,91} However, these advancements are not without challenges. High costs and extensive training requirements are significant barriers to HIT adoption, particularly in veterinary practices.^{3,11} Additionally, complex data interoperability and privacy requirements, along with staff burnout from heavy workloads and inefficient workflows, present ongoing issues.⁸ Successful strategies to address these challenges include integrating well-being metrics into performance dashboards, streamlining workflows, providing resilience training, adjusting staffing models, and using data analytics to predict and prevent burnout.¹ Applying Lean, Six Sigma, and DES methodologies enhances patient flow and resource management, while ergonomic designs improve safety and comfort.^{38,51}

Research should aim to develop cost-effective HIT solutions for smaller veterinary practices, explore advanced technologies, and investigate the longterm impacts of ergonomic interventions on patient and staff outcomes.^{9,16} Enhancing interdisciplinary collaboration is essential for sharing best practices and innovations across both human and veterinary healthcare sectors.^{39,88}

Interdisciplinary collaboration and knowledge sharing are crucial for advancing healthcare practices. By integrating insights from both human and veterinary healthcare, professionals can develop more robust and innovative solutions.

At the same time, several limitations of this review should be noted. We did not conduct a formal risk of bias assessment, certainty grading, or reporting bias analysis, as the heterogeneity and narrative nature of the included studies made these approaches impractical. Similarly, we did not systematically extract all study characteristics, such as geographic setting or funding, beyond those directly relevant to operational outcomes. These constraints, along with heterogeneity across practice types (companion, equine, and production), limit the strength of inferences, and feasibility may be constrained in resource-limited veterinary settings. This also highlights a central finding of this work: the scarcity of veterinary-specific literature underscores the need for targeted future research in this domain.

In conclusion, the methodologies and strategies explored in this paper provide a robust framework for enhancing efficiency and patient care in both human and veterinary hospitals. Our comparative approach highlighted shared challenges and fostered a better understanding of sector-specific solutions, paving the way for innovative cross-sectoral applications and continuous improvement in healthcare operations management.

However, it is important to acknowledge that differences in funding structures such as private equity, venture capital, and insurance influences can introduce bias when comparing human and veterinary healthcare systems. These financial disparities may affect operational capacity and investment in infrastructure, meaning comparisons are not always “like-for-like.”

Future research should focus on developing and validating these approaches directly in veterinary settings, particularly in areas such as client satisfaction, scheduling optimization, and resource allocation, where current veterinary evidence is limited.

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